1. We generate an initial population of random parameter vectors. We refer to each parameter vector in the population as an *individual*. Let's say this initial population has 100 individuals.
2. We iterate through this population and assess the fitness of each individual by running the model in Gridworld with that parameter vector and recording the rewards. Each individual is assigned a fitness score based on the rewards it earns. Since the initial population is random, they will all likely perform very poorly, but there will be a few, just by chance, that will perform better than others.
3. We randomly sample a pair of individuals ("parents") from the population, weighted according to their relative fitness score (individuals with higher fitness have a higher probability of being selected) to create a "breeding population."

Note

There are many different methods of selecting "parents" for the next generation. One way is to simply map a probability of selection onto each individual based on their relative fitness score, and then sample from this distribution. In this way, the most fit will be selected most often, but there will still be a small chance of poor performers being selected. This may help maintain population diversity. Another way is to simply rank all the individuals and take the top *N* individuals, and use those to mate to fill the next generation. Just about any method that preferentially selects the top performers to mate will work, but some are better than others. There's a tradeoff between selecting the best performers and reducing population diversity—this is very similar to the exploration versus exploitation tradeoff in reinforcement learning.

Figure 6.4. A genetic algorithm optimization of neural networks for reinforcement learning. A population of initial neural networks (the RL agents) is tested in the environment, earning rewards. Each individual agent is labeled by how fit it is, which is based on the rewards earned. Individuals are selected for the next generation based on their fitness; more fit individuals are more likely to be included in the next generation. The selected individuals "mate" and are "mutated" to increase genetic diversity.